

[VJC/2025/Q4]

4 Name your Jupyter Notebook as:

TASK4_<your name>_<class>_<index number>.ipynb

A student has records about bus services in Singapore stored in comma-separated value format. They want to create a suitable database to store the data and to allow them to run searches for bus route information. The database will have three tables: a table to store data about bus stops, a table about bus services, and a table about the bus routes. The fields in each table are:

Service:

- `ServiceNo` – The bus service (e.g. 118A)
- `Operator` – The bus operator company (e.g. SBST)
- `Category` – The type of route

Stop:

- `RoadName` – The road that the bus stop is on
- `Description` – A short description of the bus stop location
- `BusStopCode` – The unique code of the bus stop (e.g.75009)

Route:

- `ServiceNo` – The bus service of the route
- `Operator` – The bus operator company
- `Direction` – The direction of the route in integer representation (1 or 2 only)
- `StopSequence` – The stop's numerical sequence (First stop = 1, second stop = 2, ...)
- `BusStopCode` – The unique code of the bus stop

For each of the sub-tasks, add a comment statement at the beginning of the code using the hash symbol '#', to indicate the sub-task the program code belongs to, for example:

In [1]:

```
#Task 4.1  
Program code
```

Output:

Task 4.1

Write a Python program that uses SQL code to create the database `BUSROUTES` with the three tables given. Define the primary and foreign keys for each table. [5]

Task 4.2

The text files `Route.csv`, `Service.csv` and `Stop.csv` store the comma-separated values for each of the tables in the database.

Write a Python program to read in the data from each file and then store each item of data in the correct place in the database. [6]

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Task 4.3

Write a Python program to input a bus service number and return its routes, showing:

- the direction of the route
- the bus stop code of each stop on the route
- the road name of the bus stop
- the description of the bus stop
- in ascending order by direction and then, by stop sequence.

[7]

Save your Jupyter Notebook for Task 4.1 to 4.3.

Task 4.4

Write a Python program and the necessary files to create a web application that allows the user to input a bus stop code. The web application displays the bus service numbers that stop at that bus stop, and the corresponding bus operator. If the bus stop code is invalid, it displays an appropriate error message instead.

Save your Python program as:

TASK_4_4_<your name>_<class>_<index number>.py

with any additional files/ subfolders in a folder named:

TASK_4_4_<your name>_<class>_<index number>

[6]

Run the web application with the following inputs:

- 01111
- 01112

Save the webpage output for each input as:

TASK_4_4_<your name>_<class>_<index number>_1.html

TASK_4_4_<your name>_<class>_<index number>_2.html

[2]